

Question 1: Identify argument and form.

Consider the argument:

Alice works in Hyderabad during the week (Mon to Fri) and she stays with her parents in Karimnagar on weekends (Sat-Sun).

Therefore, Alice spends her time either in Hyderabad or in Karimnagar.

(a) [1 points] Is this a valid argument or a fallacy? Justify your answer briefly.

(b) [1 points] If it is valid, identify the argument form from the options below; otherwise, choose "None of these".

- (i) Modus Ponens $P \rightarrow Q, P \Rightarrow Q$
- (ii) Modus Tollens $P \rightarrow Q, \neg Q \Rightarrow \neg P$
- (iii) Hypothetical syllogism $P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R$
- (iv) Resolution $P \vee Q, \neg Q \Rightarrow P$
- (v) None of these

Question 2: Validity of a "wizard-muggle" style argument.

Let the domain be "people". Use suitable propositional variables.

I: If a person is a programmer, then they know how to debug code.

II: If a person is a designer, then they do not debug code.

III: Y is a designer.

IV: Therefore, Y is not a programmer.

(a) [1 points] Translate the above argument into propositional logic using implication(s). (1M)

(b) [2 points] Using either a truth table or a proof using rules of inference, determine whether the argument is a valid argument form. Clearly justify. (2M)

Question 3: Tautology check using truth tables.

Let p and q be propositional variables. Consider the formulae:

$$S_1: ((p \rightarrow q) \wedge p) \rightarrow q$$

$$S_2: q \rightarrow (p \rightarrow (p \wedge q))$$

(a) [2 points] Using truth tables, determine whether S_1 is a tautology or not. Justify from the table.

(b) [2 points] Using truth tables, determine whether S_2 is a tautology or not. Justify from the table. (Show the full truth table(s), but you may organize them neatly to save time.)

Question 4: [4 points] Logical equivalence via truth tables.

The statement $(\neg p) \rightarrow q$ is given.

Using truth tables, determine which of the following formulae are logically equivalent to $(\neg p) \rightarrow q$. Clearly state "Equivalent" or "Not equivalent" for each, with justification from the table.

~~I.~~ $p \rightarrow q$

~~II.~~ $q \rightarrow \neg p$

~~III.~~ $(\neg q) \vee \neg p$

~~IV.~~ $(p \wedge \neg q)$

(IV) $p \wedge \neg q$

Question 5: Symmetry of Lift for association rules.

You are given an association rule $X \rightarrow Y$ in a transactional database.

- (a) [1 points] Using the definition of support and confidence based on empirical probabilities, write the formula for $Lift(X \rightarrow Y)$.
- (b) [2 points] Show algebraically that $Lift(X \rightarrow Y) = Lift(Y \rightarrow X)$. Clearly show each step.

✓ 6. Apriori algorithm on a small database.

Consider the following transactional database. Items are represented by integers.

Tx id Items

1 1, 2, 3

2 1, 3, 4

3 2, 3, 4

4 1, 2, 4

5 1, 2, 3, 4

Use the Apriori algorithm with **minimum support = 2 transactions** (absolute count).

- ✓ (a) [3 points] Show the construction of frequent 1-itemsets, 2-itemsets, and (if any) 3-itemsets using the generate-and-prune steps of Apriori. Clearly show support counts at each level. (3M)
- ✓ (b) [1 points] List all frequent itemsets returned by Apriori for this database with their support counts. (1M)

Question 7: Please answer the following questions:

- a) [2+1 points] What is Levinthal's Paradox ? How does AlphaFold help to solve this paradox ?

? ~~b)~~ **[1+2 points]** Name 2 feature selection methods ? Please present 3-4 lines description for each of the methods ?

?? ~~c)~~ [1.5+1.5 points] What is the goal of Power transformation ? Please present a diagram with some description that conveys the essence of your explanation ?

Q) [1+2 points] What does dimensionality reduction mean? Please explain how it can be used to reduce overfitting?

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Question 8: Please answer the following questions:

- a) [2 points] What is supervised learning? How is it different from unsupervised learning model?

b) [2 points] Please provide two methods or algorithms for each of unsupervised and supervised learning models?

Supervised Learning	Unsupervised Learning

c) [4 points] Given that both are learning models, the concept of MSE should possibly be applicable to both – please explain in detail why or why not applicable for each of the models? If MSE is not applicable for any of the two, how would an algorithm/method that falls under the model learn?

Question 9: You are training an ML model based on Bubble sort to perform sorting in ascending order. However, you wrote a ML training program that sorts in descending order. You tested on the following vectors of numbers:

$\langle 8, 4, 9, 1, 2 \rangle$

$\langle 4, 3, 7, 2, 9 \rangle$

$\langle 1, 6, 3, 9, 7 \rangle$

$\langle 8, 5, 3, 6, 9 \rangle$

1 a) [2+2 points] Assuming your model is accurate for descending order, please compute the Bias and Variance of your program.

7 b) [2+2 points] What would be the Bias and Variance of the model if the actual order we need (i.e., the right order) is Descending order? Please provide all steps.

- c) [4 points] As extension to information in (b), Gaussian noise with a mean of 0 and variance of 2 is introduced into the vectors. What would be the MSE of the model if the actual order we need is Descending order? Please provide all steps.

Question 10: We train an ML model for regression. The training and validation Mean Squared Errors (MSE) for every X steps is given below (referred to as stepX where X can be some reasonable number):

StepX Training MSE Validation MSE

1	18.0	20.0
2	14.5	15.8
3	11.0	12.4
4	8.7	9.8
5	7.0	8.5
6	6.0	8.0
7	5.3	7.8
8	4.8	7.7
9	4.2	7.9
10	3.9	8.2
11	3.6	8.6

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Please answer the following questions:

2. [2 points] At which step X is the best model obtained?

✓
b. [2 points] Can you compute Bias for the model? Please provide steps of the computation if yes or the detailed reasoning for why not able to?

- c. [2 points] What is the final validation MSE of the selected model along with steps or reasoning involved?

d. [2 points] Please explain why training MSE continues to decrease while validation MSE increases.